

Reconstruction of the Collatz-type Map ($3n + 5$ Problem) via Jacobsthal Normalized Spinor Representation

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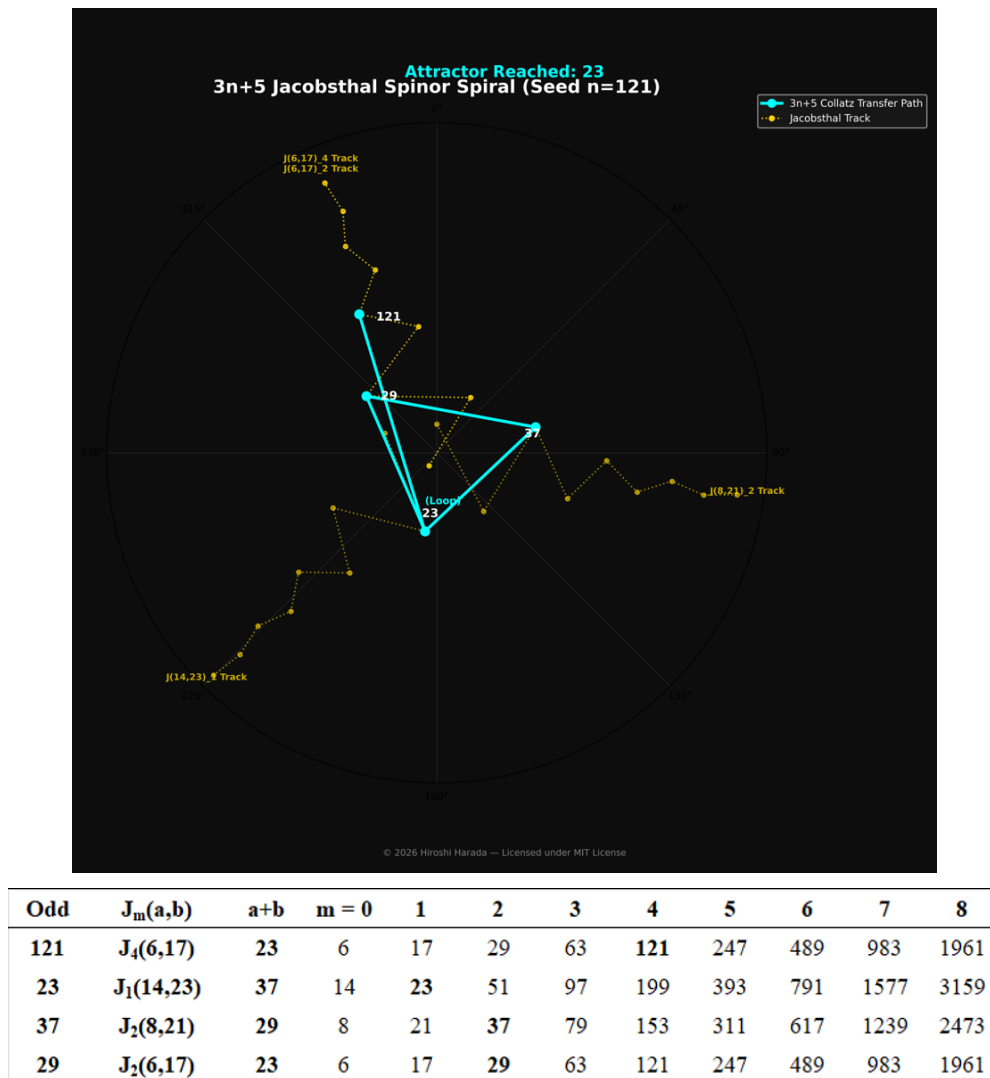


Figure 1. Visualization of the J-spinor Trajectory in Logarithmic Spiral Space.

Logarithmic spiral mapping of the odd-nucleus trajectory (cyan) and Jacobsthal (J) tracks (gold) for an initial value of $n = 121$. The visualization demonstrates the geometric topology of the sequence entering the attractor "23" from an external portal (121) via a deep spatial shift ($c = 4$) and subsequently closing as a localized standing wave.

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